CEMRACS 2018: Horizontal Gene Transfer

Vincent Calvez [1] Susely Figueroa Iglesias [2] Hélène Hivert [3]
Sylvie Méléard[4] Anna Melnykova [5] Samuel Nordmann[6]

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[1]CNRS & Institut Camille Jordan, Lyon. Vincent.Calvez@math.cnrs.fr
[2]Univ. Toulouse, IMT, UPS. Susely.Figueroa@math.univ-toulouse.fr
[3]Univ. Lyon, École centrale de Lyon. helene.hivert@ec-lyon.fr
[4]Ecole Polytechnique, France. sylvie.meleard@polytechnique.edu
[5]Univ. Cergy-Pontoise & Univ. Grenoble Alpes. anna.melnykova@u-cergy.fr
[6]CAMS, EHESS, PSL Université, Paris. samuel.nordmann@ehess.fr

Motivation

Horizontal Gene Transfer (HGT) — exchange of genetic material between unicellular/multicellular organisms not through a vertical transmission.

- the evolution of organisms
- transmission of pathogenes
- resistance to antibiotics in bacteria

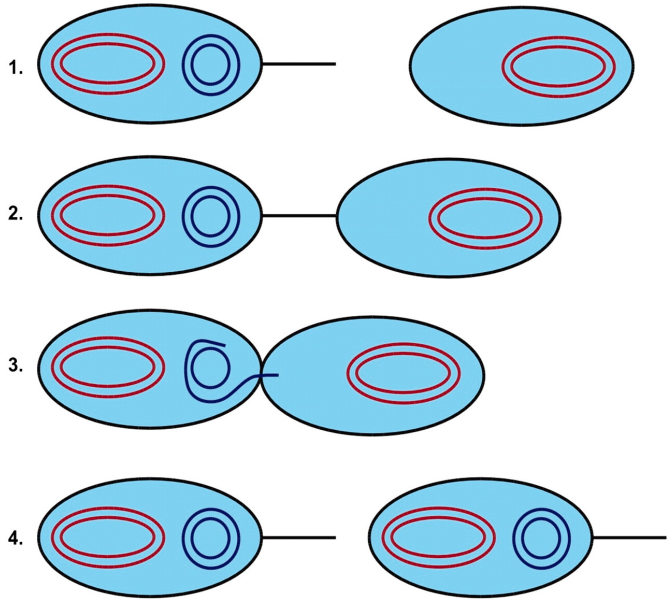
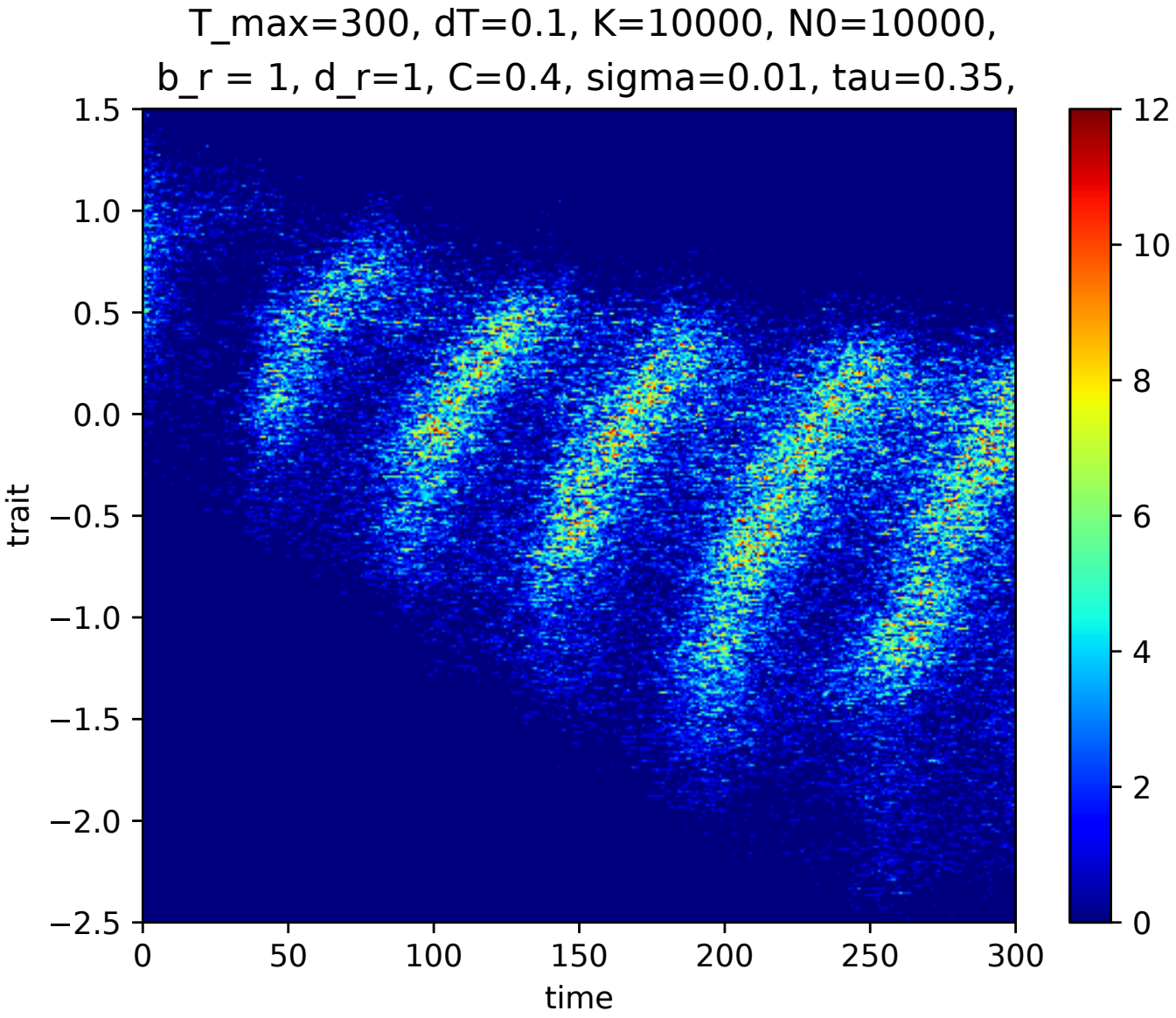


Figure 1: Unilateral HGT. Picture from Raz, Y. & Tannenbaum, E. (2010). The Influence of Horizontal Gene Transfer on the Mean Fitness of Unicellular Populations in Static Environments. Genetics, 185(1), 327–337.

Motivation: evolutionary rescue



Model

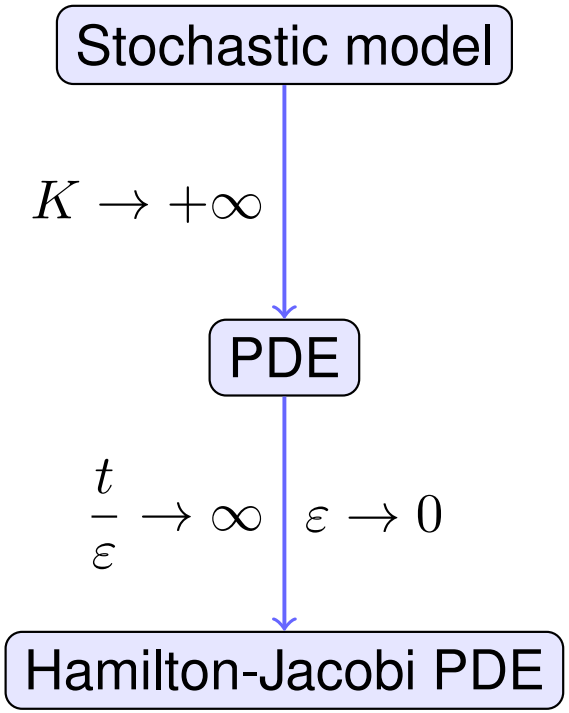
The demography of the population is regulated by:

- $X_i(t) \in \mathbb{R}^d$ is **the trait** of i-th individual living at t
- K: **scale parameter** (*carrying capacity*), N^t — size of the population at time t
- **Birth:** $b(x)$ with a mutation kernel **m(x,y).**
- **Death:** $d(x) + C\frac{N^t}{K}$**, where** C **is a competition rate**
- **Horizontal Gene Transfer (HGT):**

$$h_K(x, y, N^t) = \tau_0 \alpha(x - y) \frac{K}{N^t}$$

where α a smooth bounded function (for example: **heaviside**, **arctan**, etc..).

Structure of project



Objective:

- Grasp the evolutionary rescue with the help of a deterministic model
- Develop a stable scheme to deal with asymptotically vanishing values
- Analyze the setting which leads to such a rescue both numerically and theoretically

Stochastic model

For a population described by point measure $\nu^t(dx) = \frac{1}{K}\sum_{i=0}^{N^t} \delta_{X_i(t)}(dx)$ the generator of the process is:

$$L^K F(\nu) = \sum_{i=1}^{N} b(x_i) \int_{\mathbb{R}^d} \left(F\left(\nu + \frac{1}{K}\delta_y\right) - F(\nu)\right) m(x_i, dy)$$
$$+ \sum_{i=1}^{N} \left(d(x_i) + C\frac{N}{K}\right) \left(F\left(\nu - \frac{1}{K}\delta_{x_i}\right) - F(\nu)\right)$$
$$+ \sum_{i,j=1}^{N} h_K(x_i, x_j, \nu) \left(F\left(\nu + \frac{1}{K}\delta_{x_i} - \frac{1}{K}\delta_{x_j}\right) - F(\nu)\right).$$

Further reading:

Billiard et al. (2016a, 2015)

Stochastic simulations: algorithm

Algorithm of simulation on $[0, \Delta, \ldots, i\Delta, \ldots, N\Delta = T]$:

0 Randomly initialize the population $X^0 := \mathcal{N}(x^0_{mean}, \sigma^0) \times N^0$

1 Set: $b \equiv const$, $d = d_r x^2 + C\dfrac{N^t}{K}$, $\alpha(x) = heav(x)$

2 For $i \in [1, N-1]$:

(i) Copy X_{i-1}−th population to X_i

(ii) Compute exponential arrival times for each individual:

$$T_b := \lambda(b),\ T_d := \lambda(d),\ T_{HGT} := \lambda\left(\sum_{y \in X^i} h_K(x, y, N^{i-1})\right)$$

(iii) $\forall x \in X_i$:

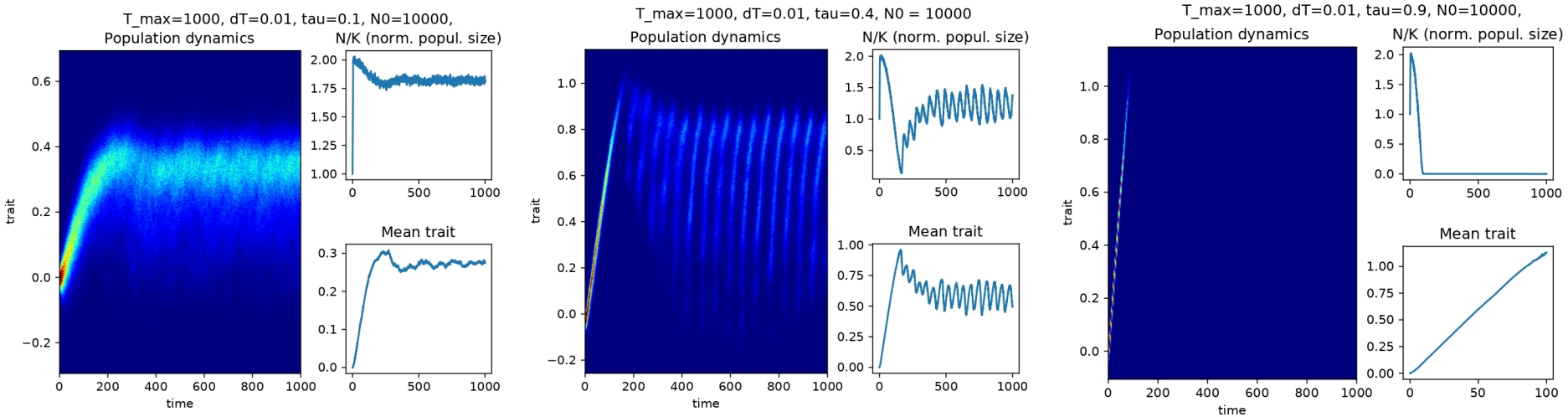
* **If $\mathbf{T_d} \leq \mathbf{\Delta}$**: remove x-th individual
* **If $\mathbf{T_b} \leq \mathbf{\Delta}$**: add a new individual with a trait from $\mathcal{N}(x, \sigma^2)$
* **If $\mathbf{T_{HGT}} \leq \mathbf{\Delta}$**: pick a trait $y \in X^i$ according to the law

$$\frac{h_K(x, y, N^{i-1})}{\sum_{y \in X^i} h_K(x, y, N^{i-1})}$$

then remove individual with trait x and add a new individual with trait y.

(iv) Return X_i

Stochastic simulations



(a) Stabilization: $\tau = 0.1$ (b) Cycles: $\tau = 0.4$ (c) Extinction: $\tau = 0.9$

Figure 2: Behavior of the population dynamics as the mutation rate τ is changing, ($b_r = d_r = 1$, $\sigma = 0.01, T_{max} = 1000, \Delta = 0.01$).

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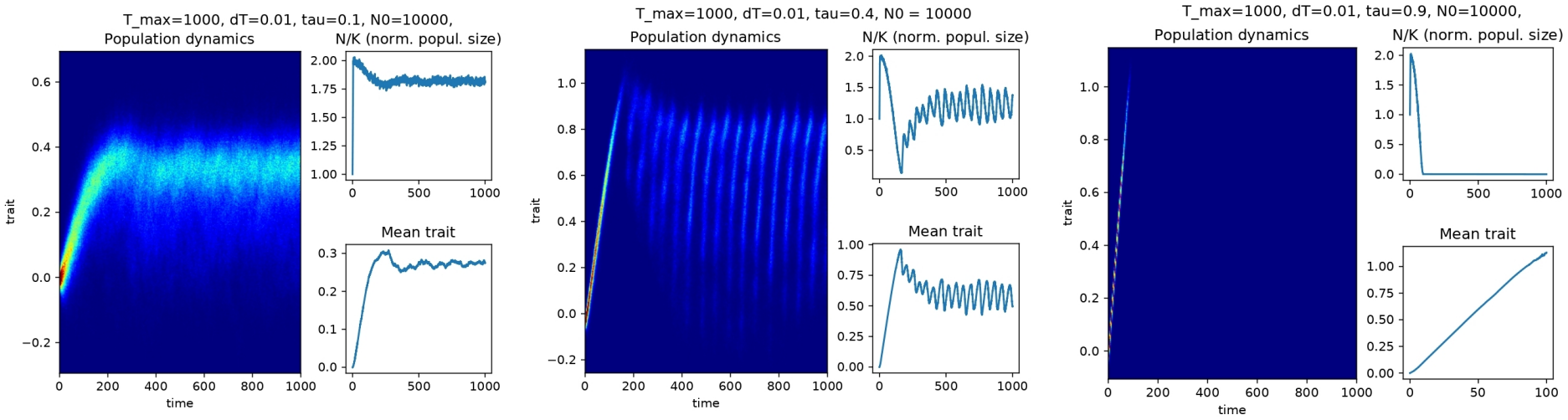

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All codes and more pictures:

`https://github.com/melnyashka/horizontal_gene_transfer`

PDE: Model and notations

The sequence of stochastic processes gives a macroscopic approximation when $K \to +\infty$.

$$\begin{cases} \partial_t f(t,x) &= -(d(x) + C\rho(t))f(t,x) + \int_{\mathbb{R}^d} m(x-y)b(y)f(t,y)dy \\ & \quad + f(t,x)\int_{\mathbb{R}^d} \tau(x-y)\frac{f(t,y)}{\rho(t)}dy, \\ & \quad \rho(t) = \int_{\mathbb{R}^d} f(t,x)dx, \qquad f(0,x) = f^0(x) > 0. \end{cases}$$

- $f(t,x) := \lim_{K\to+\infty} \nu^K$: density of the population at time t with trait x,
- $b(x), d(x), C$: birth, death and competition rate,
- $m(x-y)$: mutation kernel,
- $\tau(y-x) := \tau_0\left[\alpha(x-y) - \alpha(y-x)\right]$: HT kernel

Further reading:

Billiard et al. (2016b,a), Ferrière and Tran (2009)

PDE simulations

Explicit Euler scheme, (the integrals are computed with a left-point quadrature rule).

- Competition term: $\Delta x \sum_{j=0}^{N_x-1} f_j^n$

$$\frac{f_i^{n+1} - f_i^n}{\Delta t} = \left(- \rho^n - D_i + T_i^n \right) f_i^n + B_i^n$$

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- Transfer rate term: $\Delta x \sum_{j=0}^{N_x-1} \tau(x_i - x_j) \frac{f_j^n}{\rho^n}$

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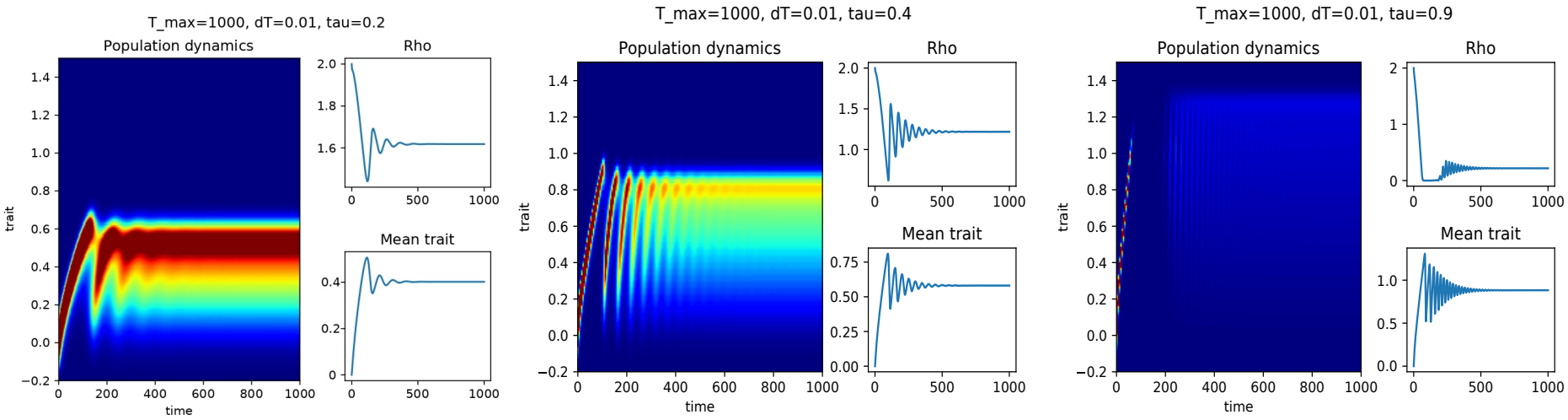
- Transfer rate term: $\Delta x \sum_{j=0}^{N_x-1} \tau(x_i - x_j) \frac{f_j^n}{\rho^n}$
- Birth rate term: $[m * (bf)]_i^n = \Delta z \sum_{k=0}^{N_z-1} m(z_k) b(x_i + z_k) f^n(x_i + z_k).$

PDE simulations



(a) Stabilization: $\tau = 0.2$ (b) Cycles: $\tau = 0.4$ (c) Regrowth: $\tau = 0.9$

Figure 3: Behavior of the population dynamics as the mutation rate τ is changing.

Shape of solutions

- Competition $\longrightarrow$ Saturation: total population $\rho(t)$ is bounded.

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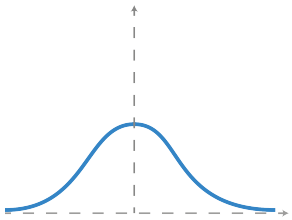
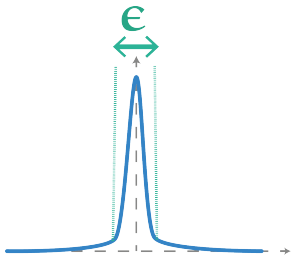
- **Small** Mutations. Rescaling parameter: $\varepsilon > 0$.

$m(y-y')$:



$\longleftrightarrow \quad \frac{1}{\varepsilon} m\left(\frac{y-y'}{\varepsilon}\right)$:



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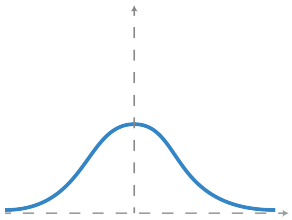

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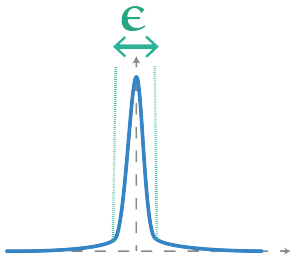

- Counterpart: **large** time

$$t \longleftrightarrow \frac{t}{\varepsilon}$$

Ecological scale $\qquad$ Evolutionary scale

$$\partial_t \longleftrightarrow \varepsilon \partial_t$$

PDE: The rescaled model

$$\begin{cases} \varepsilon \partial_t f_\varepsilon(t,x) &= -(d(x) + C\rho_\varepsilon(t)) f_\varepsilon(t,x) + \int_{\mathbb{R}} \overbrace{m(z) b(x+\varepsilon z) f_\varepsilon(t, x+\varepsilon z) dz}^{z = \frac{x-y}{\varepsilon}} \\ & \quad + f_\varepsilon(t,x) \int_{\mathbb{R}} \tau(x-y) \frac{f_\varepsilon(t,y)}{\rho_\varepsilon(t)} dy, \\ \rho_\varepsilon(t) &= \int_{\mathbb{R}} f_\varepsilon(t,x) dx, \qquad \text{+ init. conditions.} \end{cases}$$

Hopf-Cole transformation

$$f_\varepsilon \approx \text{Gaussian of variance } \varepsilon.$$

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$$\Downarrow$$

$$\partial_t u_\varepsilon = -(d(x) + \rho_\varepsilon(t)) + \int_{\mathbb{R}^d} m(z) b(x+\varepsilon z) e^{\frac{u_\varepsilon(t,x+\varepsilon z) - u_\varepsilon(t,x)}{\varepsilon}} dz + \int_{\mathbb{R}} \tau(x-y) \frac{f_\varepsilon(t,y)}{\rho_\varepsilon(t)} dy.$$

The limiting Hamilton-Jacobi equation

- $\boxed{f_\varepsilon(t,x) = e^{\frac{u_\varepsilon(t,x)}{\varepsilon}}.}$
- (Formally) $u_\varepsilon \underset{\varepsilon \to 0}{\longrightarrow} u$ (viscosity) solution of:

$$\partial_t u = -(d(x) + \rho(t)) + b(x) \int_{\mathbb{R}} M(z) e^{z \cdot \nabla_x u} dz + \tau(x - \overline{x}(t))$$

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- $\sup_{x\in\mathbb{R}} u(t,\cdot) \leq 0,$ with " $<$ "$\Leftrightarrow$ extinction.
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- Non-standard drift term

Hamilton-Jacobi: convenient framework for concentration phenomenon.
Formally :

- $\rho(t) = \max(0, b - d\bar{x}(t)^2)$.
- Adaptive dynamics:

$$\frac{\mathrm{d}}{\mathrm{d}t}\bar{x}(t) = \frac{1}{-\partial_x^2 u(t, \bar{x}(t))}\left[b - d\bar{x}(t)^2\right] \quad \text{+ possible jumps}$$

- Evolutionary Equilibrium:

$$\bar{x}_\star = \frac{\tau'(0)}{2d}.$$

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- Threshold:

$$\text{Oscillations} \quad \text{iff} \quad \frac{\tau'(0)^2}{\|\tau\|_\infty} \geq 4d.$$

- Threshold:

$$\text{Extinction} \quad \text{"iff"} \quad \|\tau\|_\infty \geq 2\sqrt{bd}.$$

Hamilton-Jacobi simulations

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Asymptotic Preserving (AP) scheme (Jin, 1999, Klar, 1998, 1999)

- avoid the increase of computational cost,
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$$\begin{tikzcd}
P_\varepsilon \arrow[r, "\varepsilon\to 0"] & P_0 \\
S_\varepsilon^h \arrow[u, "h\to 0"] \arrow[r, "\varepsilon\to 0"] & S_0^h \arrow[u, "h\to 0"']
\end{tikzcd}$$

Hamilton-Jacobi simulations: numerical AP scheme

For $u(t,x) = \varepsilon \ln f(t,x)$ we compute:

- Competition term (Implicit). Recall: $\rho^{n+1} = \sum_i e^{\frac{u_i^{n+1}}{\varepsilon}}$.

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- Death rate term

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- Birth rate term
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Remarking that for the exponential computations we make

$$\frac{f^\varepsilon(t^n,y)}{\rho^\varepsilon(t^n)} = \frac{\mathrm{e}^{u^\varepsilon(t^n,y)/\varepsilon}}{\int_{\mathbb{R}} \mathrm{e}^{u^\varepsilon(t^n,z)/\varepsilon}\mathrm{d}z} = \frac{\mathrm{e}^{(u^\varepsilon(t^n,y)-\max_x u^\varepsilon(t^n,x))/\varepsilon}}{\int_{\mathbb{R}} \mathrm{e}^{(u^\varepsilon(t^n,z)-\max_x u^\varepsilon(t^n,x))/\varepsilon}\mathrm{d}z},$$

Hamilton-Jacobi simulations (PDE for $\varepsilon << 1$):

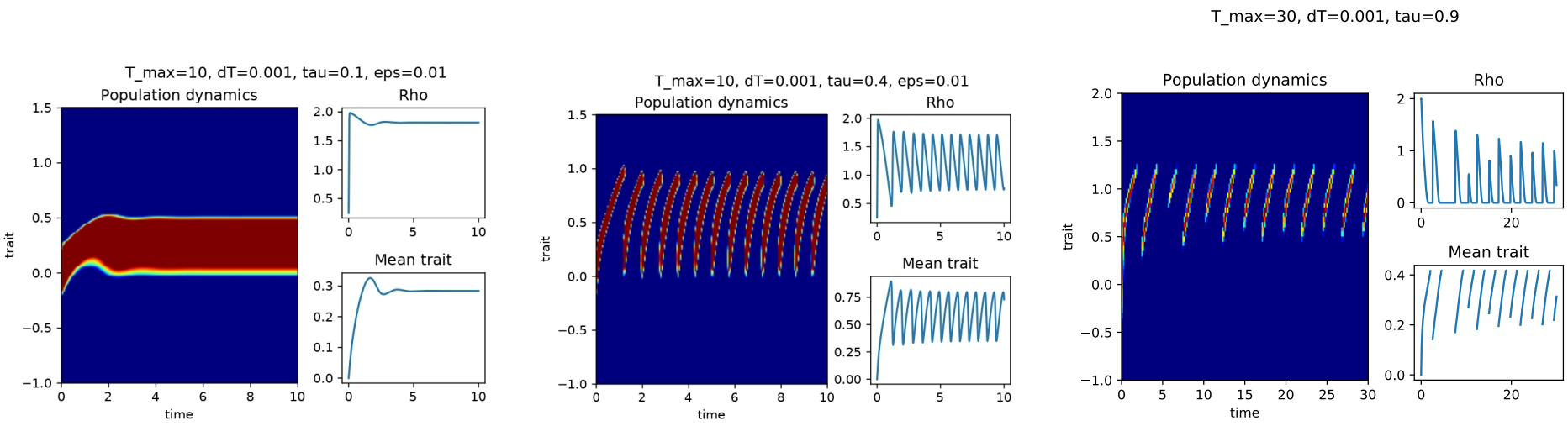

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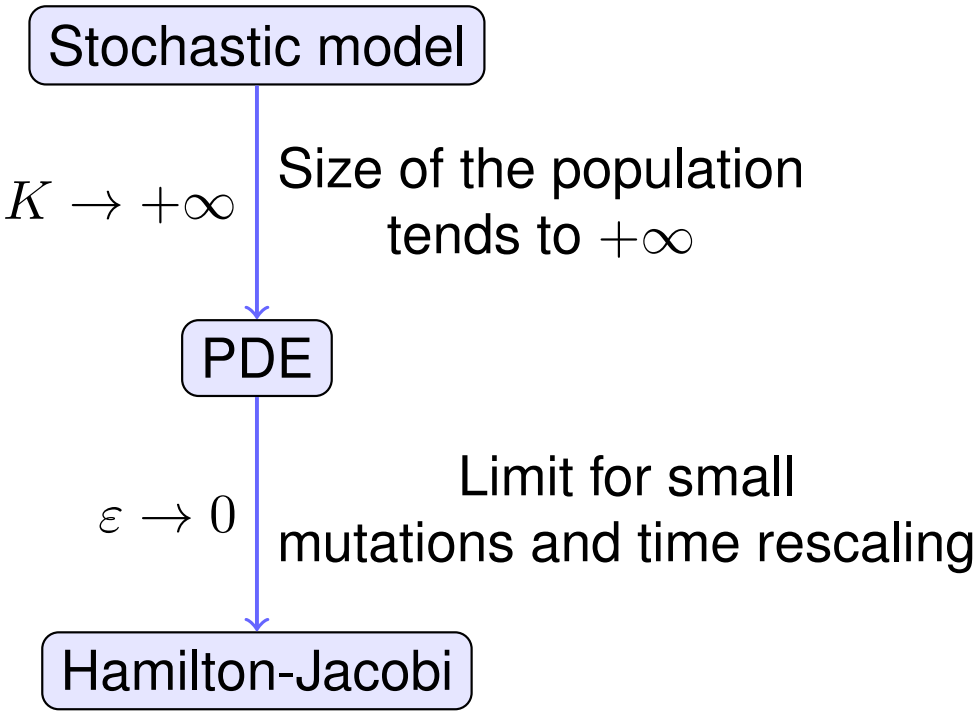
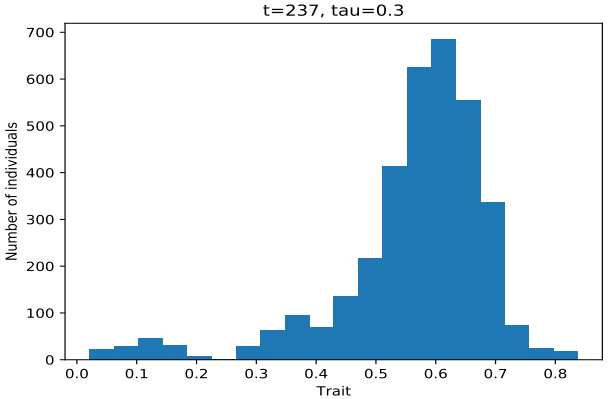
Figure 4: Behavior of the population dynamics as the mutation rate τ is changing.

Conclusion

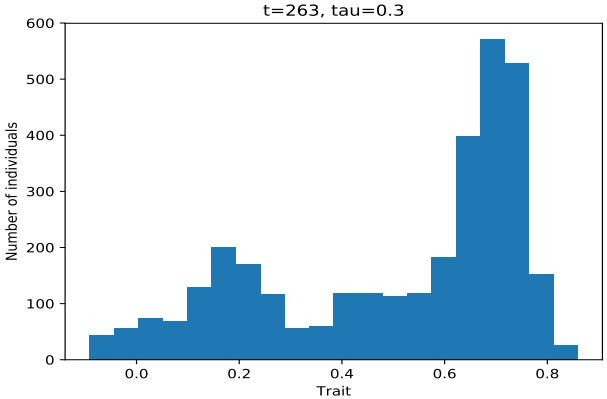


- After two limiting procedures, Hamilton-Jacobi model is still able to grasp the qualitative properties of the stochastic model.
- H-J is convenient to describe where the population concentrates. Our numerics shows that it is also able to describe *small population*.
- H-J framework allows a rigorous and rich description.

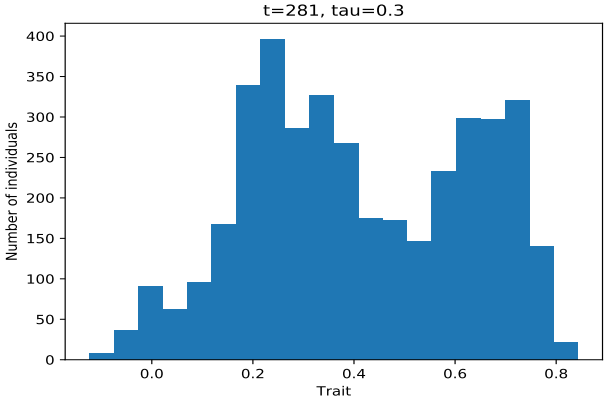
Evolutionary rescue



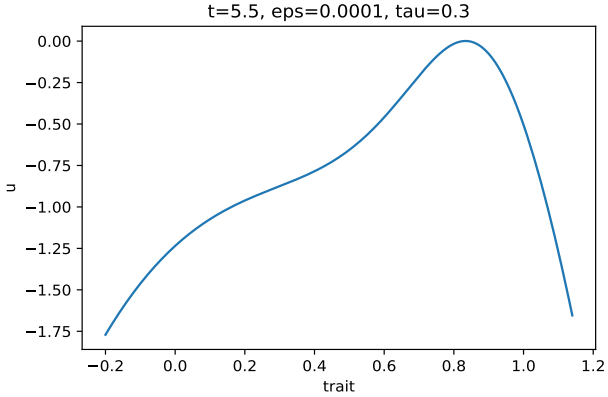
(a) $t = 237$



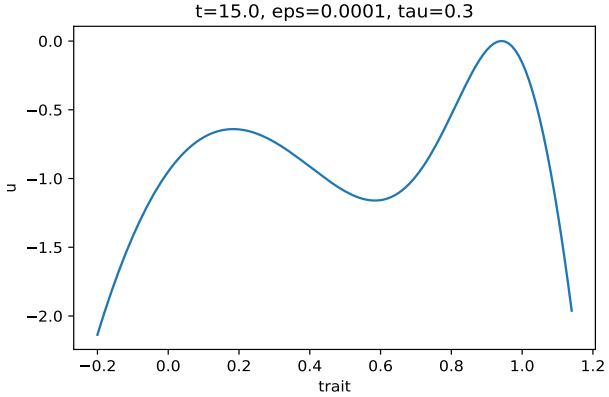
(b) $t = 263$



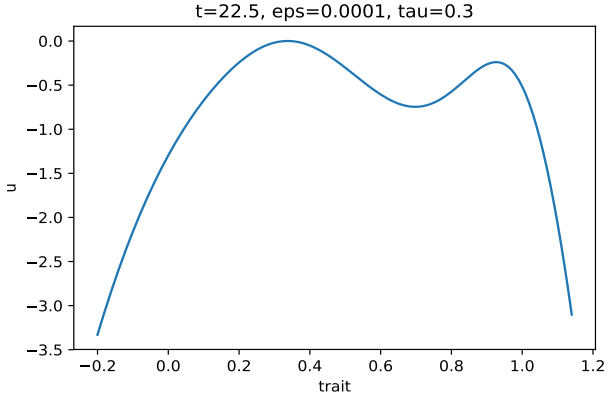
(c) $t = 281$



(d) $t = 5.5$



(e) $t = 15.0$



(f) $t = 22.5$

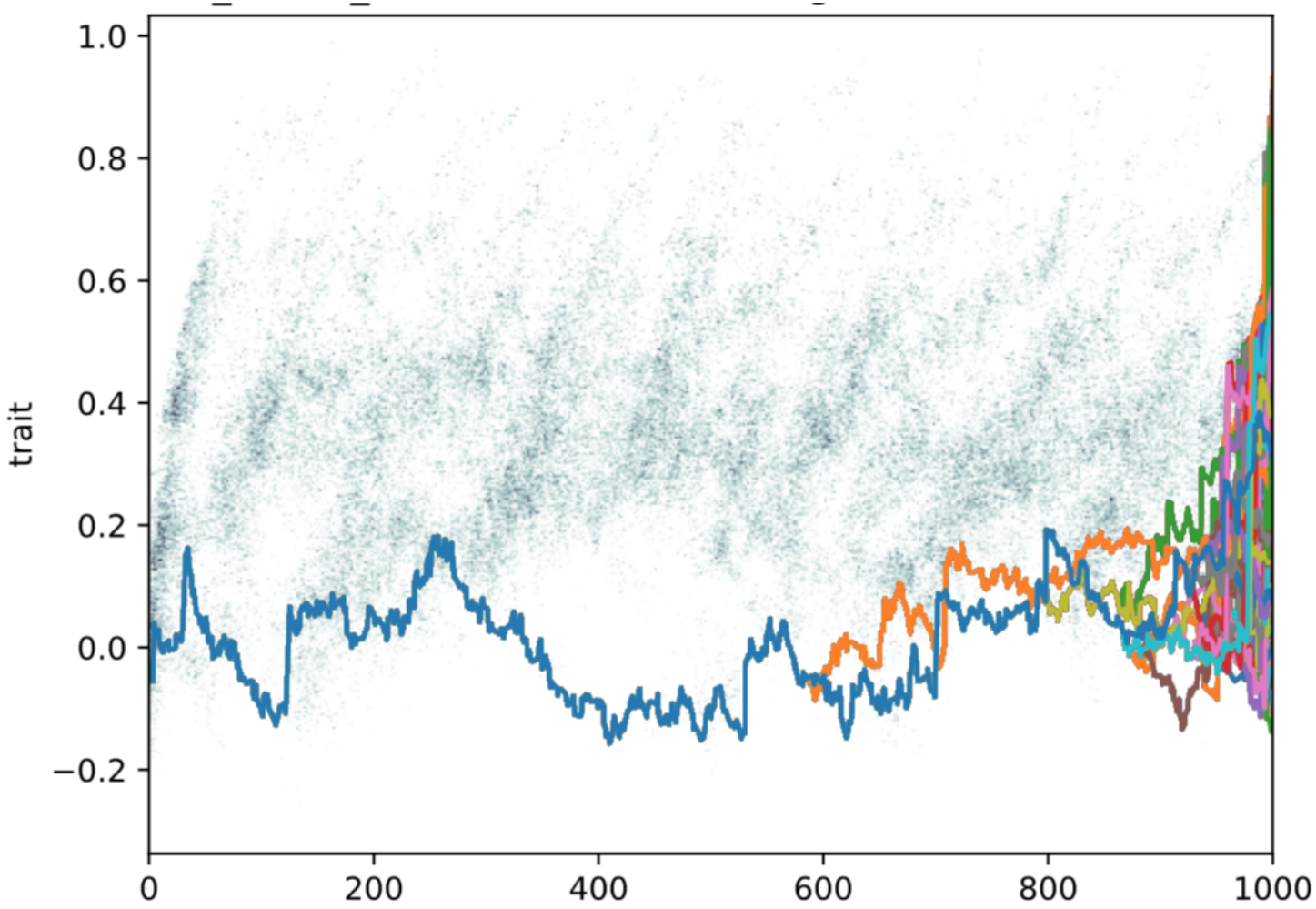
Figure 5: Stochastic/HJ. $\varepsilon = 10^{-4}$ and $\tau = 0.3$

Open questions

1. Rigorous proofs.
2. Formula for the period of oscillations.

Open questions

1. Rigorous proofs.
2. Formula for the period of oscillations.
3. Reconstruction of lineages with the characteristics of Hamilton-Jacobi.



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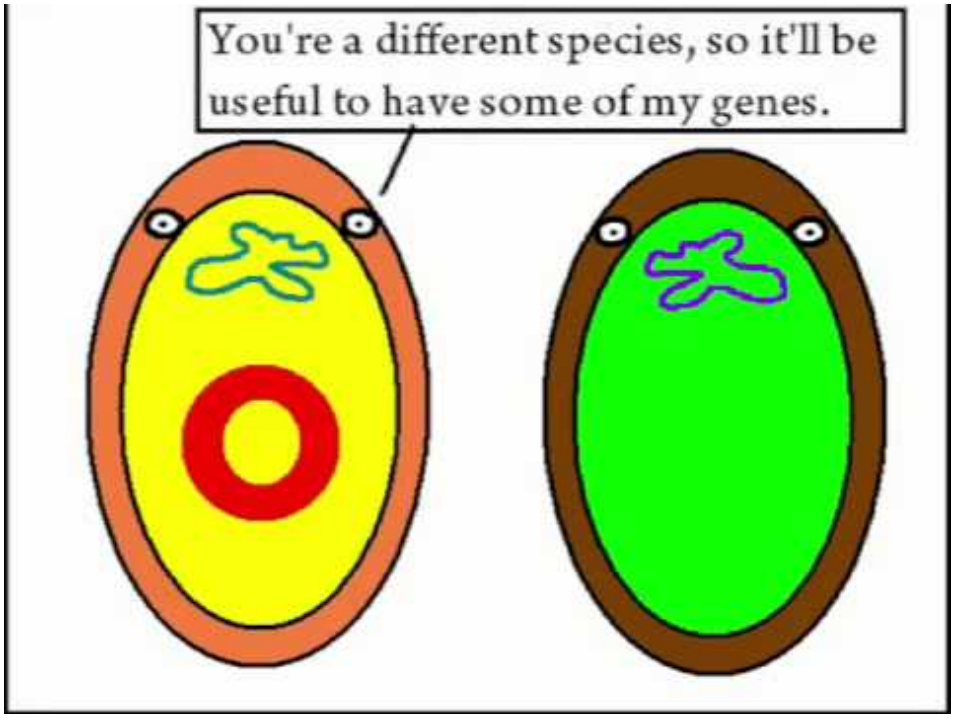
THANK YOU!

MERCI!

GRACIAS!

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